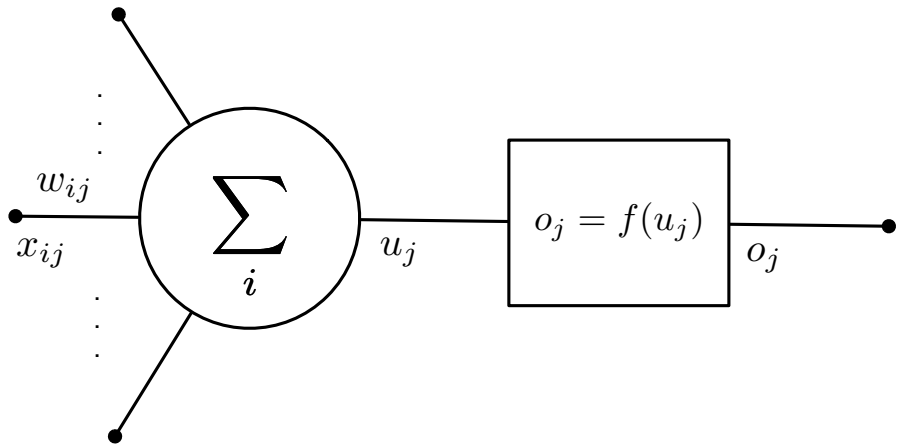
1. DotProduct unit in NNS library

For unit *j :*

Input: x_{ij}

Output: o_j

Weights: w_{ij}



Status:

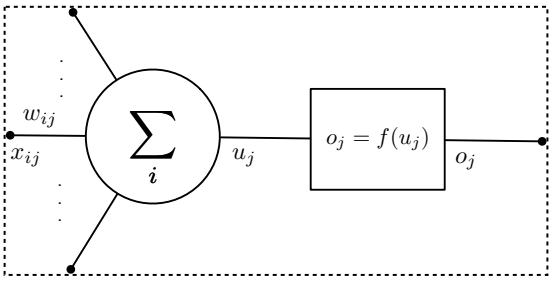
1.1. $u_j = \sum_i w_{ij} x_{ij}$

Output:

1.2. $o_j = f(u_j)$

1.3. $f'(o_j) = \dfrac{df}{du_j}$

2. Standard Error Backpropagation



The aim is to find a set of weights $[w_{ij}]$ minimising the error function:

2.1. $$E = \frac{1}{N_B}\sum_{p=0}^{N_B} E_p(w_{ij}, x_p, t_p)$$

where $E_p(w_{ij}, x_p, t_p)$ id the error the *p-th* example $[x_p, t_p]$ with x_p input pattern and t_p related desired output. Using gradient descent, we can iteratively update the weights according to the following eq 2.2. This will lead eventually to a local minimum of E.

2.2. $$\Delta w_{ij} = -\epsilon \frac{\partial E}{\partial w_{ij}} = -\epsilon \sum_p \frac{\partial E_p}{\partial w_{ij}}$$

2.3. $$\frac{\partial E_p}{\partial w_{ij}} = \frac{\partial E_p}{\partial u_j^p}\frac{\partial u_j^p}{\partial w_{ij}} = \frac{\partial E_p}{\partial o_j^p}\frac{\partial o_j^p}{\partial u_j^p}\frac{\partial u_j^p}{\partial w_{ij}} = \frac{\partial E_p}{\partial o_j^p} f'(u_j^p)\frac{\partial u_j^p}{\partial w_{ij}}$$

where:

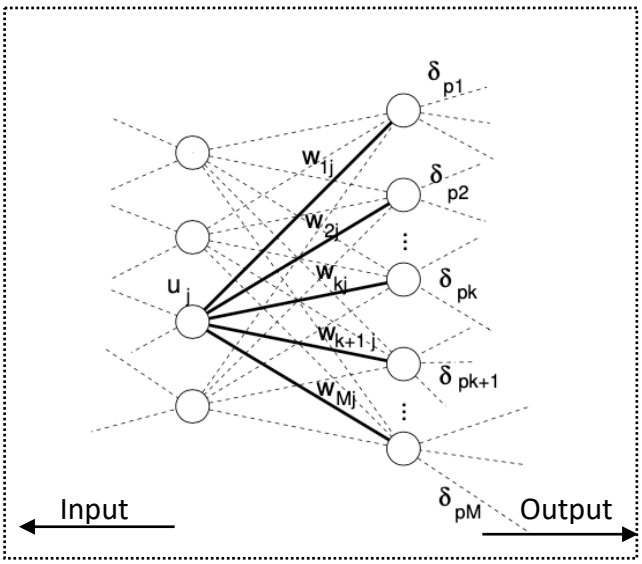
2.4. $$\frac{\partial u_j^p}{\partial w_{ij}} = \frac{\partial}{\partial w_{ij}}\left(\sum_k w_{kj} o_k^p\right) = \frac{\partial}{\partial w_{ij}} w_{ij} o_i^p = o_i^p$$

For output units and MSE loss:

2.5. $$\frac{\partial E_p}{\partial o_j^p} = o_j^p - t_j^p$$

2.6. $$\frac{\partial E_p}{\partial w_{ij}} = (o_j^p - t_j^p)\ f'(u_j^p)\ o_j^p$$

For hidden units, considering E as a function of the status of units receiving input from neuron j, we obtain:



$$\frac{\partial E_p}{\partial o_j^p} = \sum_k \frac{\partial E_p}{\partial u_k^p}\frac{\partial u_k^p}{\partial o_j^p} = \sum_k \frac{\partial E_p}{\partial o_k^p}\frac{\partial o_k^p}{\partial u_k^p}\frac{\partial u_k^p}{\partial o_j^p} =$$

2.7.
$$= \sum_k \frac{\partial E_p}{\partial o_k^p}\frac{\partial o_k^p}{\partial u_k^p}\frac{\partial}{\partial o_j^p}\sum_i w_{ik} o_i^p =$$

$$= \sum_k \frac{\partial E_p}{\partial o_k^p}\frac{\partial o_k^p}{\partial u_k^p} w_{jk} = \sum_k \frac{\partial E_p}{\partial o_k^p} f'(u_k^p) w_{jk}$$

2.8.
$$\frac{\partial E_p}{\partial w_{ij}^p} = \frac{\partial E_p}{\partial o_j^p}\frac{\partial o_j^p}{\partial u_j^p}\frac{\partial u_j^p}{\partial w_{ij}} = \frac{\partial E_p}{\partial o_j^p}\frac{\partial o_j^p}{\partial u_j^p} o_i^p$$

$$\frac{\partial E_p}{\partial w_{ij}} = o_i^p \delta_j^p$$

with:

2.9. $$\delta_j^p = \frac{\partial E_p}{\partial u_j^p} = \frac{\partial E_p}{\partial o_j^p}\frac{\partial o_j^p}{\partial u_j^p} = \begin{cases} \frac{\partial E_p}{\partial o_j^p} f'(u_j^p) & if\ j \text{ is output} \\ \left(\sum_k w_{jk}\delta_k^p\right) f'(u_j^p) & if\ j \text{ is hidden} \end{cases}$$

For a batch of N_B patterns, the network weights are updated according to:

2.10. $$\Delta w_{ij} = -\epsilon \frac{\partial E}{\partial w_{ij}} = -\epsilon\ \frac{1}{N_B}\sum_p o_i^p\ \delta_j^p \qquad \left(E = \frac{1}{N_B}\sum_p E_p\right)$$

3. Error Back-propagation with regularisation

Excessive reduction of training error (over-training) is prone to increase testing error so reducing the generalisation capabilities of a neural network. The network essentially memorises the training cases, which hinders its ability to handle new cases (e.g., the test set).

In general a neural network with low training error (bias) has trouble in reducing the test error (variance). The bias is defined as:

$$Bias(\tilde{\theta}_m) = E(\tilde{\theta}_m) - \theta$$

Where θ is the parameterisation of the true generating function, and $\tilde{\theta}_m$ is the parameterisation of an estimator based on the sample m. In practice, $\tilde{\theta}_m$ are weights of the network after training on a sample set m. The variance is given by:

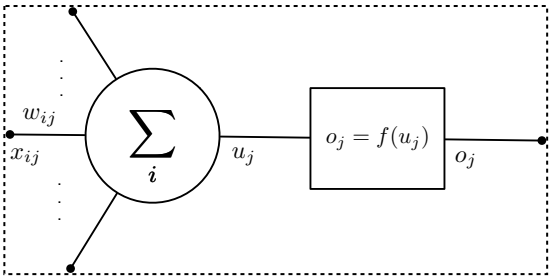
$$Var(\tilde{\theta}_m) = E[(\tilde{\theta}_m - E(\tilde{\theta}_m))^2]$$

It expresses the change of estimator's results when other data from same generator are used in estimation.

Regularisation provides an attempt to combat the bias-variance tradeoff so as to produce a $\tilde{\theta}_m$ with low bias and low variance. In this perspective, the error function is modified by adding a regularisation term. In practice, the most diffused regularisers are:

L_2 norm of weight vector $\sum_{w_{ij}} w_{ij}^2$;

L_1 norm of weight vector $\sum_{w_{ij}} \left| w_{ij} \right|$.



Where sum are over the entire set of network parameters $[w_{ij}]$ (including weights and biases) .

In the case of L_2 norm:

3.1. $$E = \frac{1}{N_B} \sum_{p=0}^{N_B} E_p(w_{ij}, x_p, t_p) + \alpha \sum_{w_{ij}} w_{ij}^2$$

where α is a parameter controlling the degree of regularisation.

3.2. $$\Delta w_{ij} = -\epsilon \frac{\partial E}{\partial w_{ij}} = -\epsilon \left(\sum_p \frac{\partial E_p}{\partial w_{ij}} + 2\alpha w_{ij} \right)$$

3.3. For output units and MSE loss, the term:

$$\frac{\partial E_p}{\partial w_{ij}}$$

is computed as in 2.5 - 2.6. We obtain:

3.4. $$\frac{\partial E_p}{\partial o_j^p} = o_j^p - t_j^p$$

3.5. $\dfrac{\partial E_p}{\partial w_{ij}} = (o_j^p - t_j^p)\ f'(u_j^p)\ o_i^p$

For a B-sized batch, the weight update rule is:

3.6. $\Delta w_{ij} = -\epsilon \dfrac{\partial E}{\partial w_{ij}} = -\epsilon \dfrac{1}{B}(\sum_p (o_j^p - t_j^p)\ f'(u_j^p)\ o_j^p + 2\alpha w_{ij})$

For hidden units, the term $\dfrac{\partial E_p}{\partial w_{ij}}$ is obtained similarly to standard BP. It must be pointed out that error back-propagation starts with 3.4 (or other value related to the adopted error function). Also in this case we have:

3.7. $\dfrac{\partial E_p}{\partial w_{ij}} = o_i^p \delta_j^p$

with:

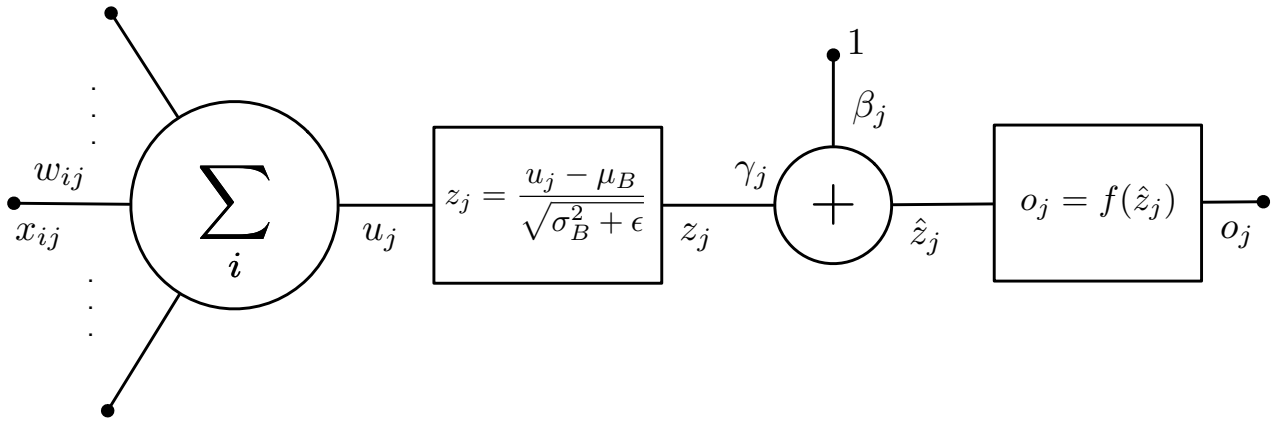
3.8. $\delta_j = \dfrac{\partial E_p}{\partial u_j^p} = \dfrac{\partial E_p}{\partial o_j^p}\dfrac{\partial o_j^p}{\partial u_j^p} = \begin{cases} \dfrac{\partial E_p}{\partial o_j^p} f'(u_j^p) & if\ j \text{ is output} \\ \left(\sum_k w_{jk}\delta_k^p\right) f'(u_j^p) & if\ j \text{ is hidden} \end{cases}$

From 3.2, for a B-sized batch, weights are updated according to:

3.9. $\Delta w_{ij} = -\epsilon \dfrac{\partial E}{\partial w_{ij}} = -\epsilon \dfrac{1}{B}\sum_p (o_i^p\ \delta_j^p + 2\alpha w_{ij})$

If L_1 regularizer is used, the derivative term $2\alpha w_{ij}$ is replaced by $sgn(w_{ij})$ in eq. 3.2, 3.6 and 3.9.

4. Batch normalization Backpropagation



BNTransform convert the status u_j into:

1. $$z_j = \frac{u_j - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}$$

Where μ_B and σ_B^2 are the batch mean and variance respectively. The activation function is applied to the new status variable $\hat{z}_i$:

2. $\hat{z}_j = \gamma z_j + \beta$
3. $o_j = f(\hat{z}_j)$

In addition to w_{ij} , for each unit γ_j and β_j must be learned along with the estimate of μ_j and σ_j. Network parameters must be updated according to the equations:

4. $$\Delta w_{ij} = -\epsilon \frac{\partial E}{\partial w_{ij}} = -\epsilon \sum_p \frac{\partial E_p}{\partial w_{ij}}$$
$$\Delta \gamma_j = -\epsilon \frac{\partial E}{\partial \gamma_j} = -\epsilon \sum_p \frac{\partial E_p}{\partial \gamma_j}$$
$$\Delta \beta_j = -\epsilon \frac{\partial E}{\partial \beta_j} = -\epsilon \sum_p \frac{\partial E_p}{\partial \beta_j}$$

where $p = 1 \ldots N_B$ being N_B the min-batch size

5. $$\frac{\partial E}{\partial \hat{z}_j} = \frac{\partial E}{\partial o_j} f'(\hat{z}_j)$$

6. $$\frac{\partial E}{\partial \beta_j} = \frac{\partial E}{\partial \hat{z}_j} = \frac{\partial E}{\partial o_j} f'(\hat{z}_j)$$

7. $$\frac{\partial E}{\partial \gamma_j} = z_j \frac{\partial E}{\partial \hat{z}_j} = z_j \frac{\partial E}{\partial o_j} f'(\hat{z}_j)$$

8. $$\boxed{\begin{aligned} \Delta\gamma_j &= -\epsilon \sum_p z_j \frac{\partial E_p}{\partial o_j} f'(\hat{z}_j) \\ \Delta\beta_j &= -\epsilon \sum_p \frac{\partial E_p}{\partial o_j} f'(\hat{z}_j) \end{aligned}}$$

9. $$\frac{\partial E}{\partial u_j} = \sum_p \frac{\partial E}{\partial z_p} \frac{\partial z_p}{\partial u_j}$$

10. $$\frac{\partial z_i}{\partial u_j} = \begin{cases} \frac{1}{\sqrt{\sigma_b^2 + \epsilon}} + \frac{-1}{B\sqrt{\sigma_b^2 + \epsilon}} + \frac{-(u_j - \mu_B)^2}{B(\sigma_b^2 + \epsilon)^{3/2}} & if \quad i = j \\ \frac{-1}{B\sqrt{\sigma_b^2 + \epsilon}} + \frac{-(u_j - \mu_B)^2}{B(\sigma_b^2 + \epsilon)^{3/2}} & if \quad i \neq j \end{cases}$$

11. $$\frac{\partial E}{\partial u_j} = \frac{1}{\sqrt{\sigma_b^2 + \epsilon}} \frac{\partial E}{\partial z_j} + \frac{-1}{B\sqrt{\sigma_b^2 + \epsilon}} \sum_i \frac{\partial E}{\partial z_i} - \frac{1}{B(\sigma_b^2 + \epsilon)^{3/2}} \sum_i \frac{\partial E}{\partial z_i} (u_j - \mu_B)^2$$

where:

12. $$\frac{\partial E}{\partial z_j} = \gamma \frac{\partial E}{\partial \hat{z}_j}$$

Finally, values $\frac{\partial E}{\partial u_j}$ obtained from equation 11. and 12. are back-propagated as in equation 2.9.